

something is not right and gives a chance to recover. If the pilot cannot fix the situation in time, the design of the plane takes over. A wing cuff on the aircraft separates the wing into two distinct parts: one that allows a stall to begin and the other that prevents it from continuing.

Edible smartpill that could be powered inside the body

In the near future, you will have a number of technologies living in your body, curing your illnesses, regrowing damaged nerves and keeping things running smoothly. However, considering most of this technology is surgically implanted, what will we do when it has finished doing its job? Enter, edible electronics!

According to Christopher Bettinger of Carnegie Mellon University, USA, edible electronics would consist of electronic materials such as circuits and batteries that would be just as edible as the food and vitamins we ingest daily. These would be powered by charged ions that reside within our own digestive tracts, and after completing their job, would dissolve within our bodies.

Edible electronics are set to enter the market soon. One such device called Ingestion Event Marker has been approved by Food and Drug Administration, USA. It provides digital feedback on health and is designed to improve patients' health habits and their connections to caregivers.

Mushrooms help power smartphones

Researchers have created a new type of lithium-ion battery anode using portabella mushrooms, which are inexpensive, environmentally friendly and easy to produce. These are unlike the current industry standard that is synthetic graphite and comes with a high cost of manufacturing because it requires tedious purification and preparation processes, which are also harmful to the environment.

According to Cengiz Ozkan, professor of mechanical engineering and materials science and engineering at University of California, USA, nanocarbon architectures derived from biological materials such as mushrooms can be considered a green and sustainable alternative to graphite based anodes.

Biomass can be used as a replacement for graphite as it has high carbon content, low cost and is environmental friendly. Researchers were drawn to using mushrooms as a form of biomass because these are highly porous.

Porosity is important for batteries because it creates more space for the storage and transfer of energy, a critical component for improving battery performance. Also, the high potassium salt concentration in mushrooms allows for increased electrolyte-active material over time by activating more pores, gradually increasing its capacity.

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